Week 13 Progress Report

Details

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- **PRN:** 22070521041
- **Name of Industry:** SS Infotech Pvt. Ltd.
- **Reporting Week: (30/03/2026 to 03/04/2026)**
- **Institute Mentor Name:** Dr. Shreyas Hole
- **Industry Mentor Name:** Mr. Viraj Patle

Overview of the Week

- Week 13 focused on writing and deploying a custom detection rule in the Wazuh SIEM to enhance the existing threat detection capabilities of the SOC environment. Rather than relying solely on built-in Wazuh rules, this week introduced the practice of custom rule authoring — a core skill in real-world SOC operations used to tune detection logic for specific threat scenarios.

- The custom rule (ID: 100002) was written in XML and placed in the dedicated custom rules file at /var/ossec/etc/rules/local_rules.xml on the Wazuh Manager (Ubuntu Server 24.04). The rule was designed to monitor the built-in Windows failed login rule (Rule ID: 60122, Event ID: 4625) and trigger a high-severity consolidated alert when 5 or more failed login attempts occurred within a 120-second window. The rule was tagged with MITRE ATT&CK technique T1078 (Valid Accounts) and assigned Level 10 (High) severity.

- After deploying the custom rule and restarting the Wazuh Manager service, the attack simulation was performed by triggering 6 intentional failed login attempts on the Windows 11 monitored endpoint using the runas command with an invalid user account in rapid succession. The Wazuh Dashboard confirmed both the individual Rule 60122 alerts (6 events) and the consolidated custom Rule 100002 alert (Level 10) firing as expected.

Objectives of the Week

- **To write a custom Wazuh detection rule (ID: 100002) in local_rules.xml that builds upon the existing Windows failed login detection (Rule 60122, Event ID 4625).**
- **To configure the custom rule with a frequency threshold of 5 failed events within a 120-second window, triggering a Level 10 (High) severity alert.**
- **To map the custom rule to MITRE ATT&CK technique T1078 (Valid Accounts) to align with industry-standard threat classification.**
- **To restart the Wazuh Manager and validate that the custom rule was loaded without errors.**
- **To simulate multiple failed login attempts on the Windows 11 endpoint and verify that both individual Rule 60122 alerts and the consolidated Rule 100002 alert fired correctly in the Wazuh Dashboard.**
- **To demonstrate the SOC analyst skill of custom rule authoring for enhanced and tuned threat detection beyond default SIEM configurations.**

Key Accomplishments

- Successfully authored and deployed a custom Wazuh detection rule (ID: 100002, Level 10) in **local_rules.xml** on the Wazuh Manager (Ubuntu Server 24.04).
- Configured rule chaining using **if_matched_sid** to build on the existing Rule 60122 (Windows Event ID 4625 — failed login), with a frequency threshold of 5 attempts within 120 seconds.

- Tagged the custom rule with MITRE ATT&CK technique **T1078 (Valid Accounts)** for proper threat classification.
- Restarted the Wazuh Manager service successfully and confirmed active status using systemctl.
- Simulated 6 failed login attempts on the Windows 11 endpoint using the **runas** command in PowerShell within a 2-minute window.
- Verified 6 individual Rule 60122 alerts on the Wazuh Dashboard confirming per-attempt detection.
- Confirmed 1 consolidated Rule 100002 (Level 10, High) alert on the Wazuh Dashboard — validating that the custom rule fired and correctly described: **"Custom Rule: Multiple failed login attempts detected on Windows endpoint (Event ID 4625)".**

Issues Faced & Resolutions – Week 13

- **1. IP address conflict between Windows 11 host and Ubuntu Server**

  Issue: When attempting to SSH into the Ubuntu Server (192.168.1.6) from Windows PowerShell to deploy the custom rule, the connection was refused despite SSH being active, UFW being inactive, and ping succeeding. Investigation revealed that the Windows 11 host's Wi-Fi adapter had been dynamically assigned the same IP address (192.168.1.6) by the router — causing the Windows machine to connect to itself rather than the Ubuntu Server.

  Resolution: Ran ipconfig /release followed by ipconfig /renew in Windows PowerShell to force the router to assign a new IP address to the Windows machine. Ubuntu Server retained its static IP (192.168.1.6) as configured via Netplan in Week 11. SSH connection from Windows to Ubuntu Server was re-established successfully without any changes to ossec.conf, the Wazuh agent configuration, or the Netplan setup.

- **2. Editing the local_rules.xml file using nano**

  Issue: When attempting to manually edit the local_rules.xml file using nano, clearing the existing default content and replacing it with the new custom rule XML proved difficult using standard keyboard navigation within the terminal.

  Resolution: Used the sudo tee command with a heredoc (EOF block) to overwrite the file contents entirely in a single terminal command. This approach bypassed the need for manual editing in nano and ensured the correct XML was written to the file without formatting errors.

Tools Used – Week 13

- **Wazuh Manager (Ubuntu Server 24.04, local_rules.xml)** — Platform used to write, deploy, and reload the custom detection rule (Rule ID 100002).
- **Wazuh Dashboard (v4.7.5)** — Used to verify alert firing by searching for rule.id: 60122 (individual events) and rule.id: 100002 (consolidated custom alert).
- **Windows PowerShell (Windows 11 Endpoint)** — Used to trigger failed login events via the runas command, and to resolve the IP conflict using ipconfig /release and /renew.
- **Ubuntu Server 24.04 Terminal (SSH)** — Used to write the custom rule XML using sudo tee, and to restart the Wazuh Manager service.
- **VirtualBox** — Hypervisor hosting the Ubuntu Server VM on the internal network.

Conclusion of Week 13

Week 13 successfully demonstrated the ability to extend Wazuh's detection capabilities through custom rule authoring — a critical SOC analyst skill that goes beyond default SIEM configurations. By writing Rule 100002 in local_rules.xml with frequency-based correlation and MITRE T1078 tagging, the SOC lab now has a tuned detection layer that consolidates multiple individual failed login alerts into a single high-severity event. This reduces alert fatigue while ensuring that sustained credential-based attack patterns are immediately surfaced with appropriate severity. The troubleshooting performed during this week — particularly resolving the IP conflict using dynamic IP renewal rather than modifying the static Netplan configuration — reinforced the

importance of systematic network debugging in a multi-VM lab environment. The lab environment continues to grow in detection maturity, with custom rules now complementing the built-in Wazuh ruleset established in earlier weeks.

Alerts Generated – Week 13

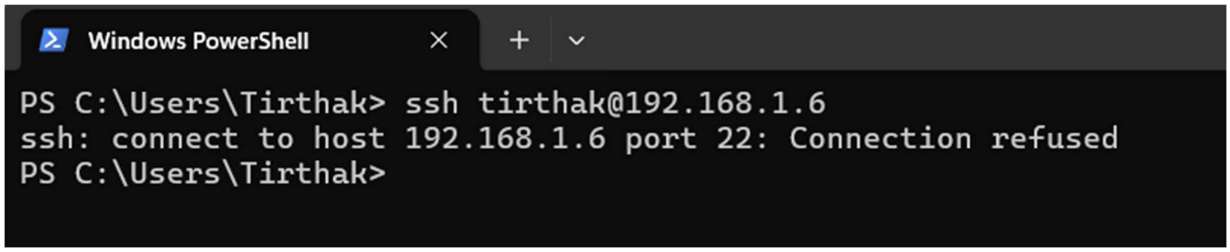
Rule ID	Description	Level
60122	Windows failed user logon (Event ID 4625)	5
100002	**Custom Rule: Multiple failed login attempts detected on Windows endpoint (Event ID 4625)**	**10 (High)**

Custom Rule 100002 – Full Details:

Field	Value
Rule ID	100002
Rule Level	10 (High)
Description	Custom Rule: Multiple failed login attempts detected on Windows endpoint (Event ID 4625)
Base Rule Monitored	60122 (Windows Event ID 4625 — Failed Logon)
Frequency Threshold	5 failed events
Time Window	120 seconds
MITRE ID	T1078
MITRE Technique	Valid Accounts
Rule File	/var/ossec/etc/rules/local_rules.xml
Log Source	Windows Security Event Log

Screenshots of Activities (if Any)

[Please insert screenshots here — custom rule XML in local_rules.xml, Wazuh Manager restart confirmation, rule.id: 60122 showing 6 individual failed login events, rule.id: 100002 showing Level 10 consolidated custom alert with description]



```
tirthak@wazuh-server:~$ sudo systemctl status ssh
[sudo] password for tirthak:
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
     Active: active (running) since Mon 2026-03-23 14:29:47 UTC; 5min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 701 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 787 (sshd)
      Tasks: 1 (limit: 6796)
     Memory: 4.4M
        CPU: 227ms
     CGroup: /system.slice/ssh.service
             └─787 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 23 14:29:46 wazuh-server systemd[1]: Starting OpenBSD Secure Shell server...
Mar 23 14:29:47 wazuh-server sshd[787]: Server listening on 0.0.0.0 port 22.
Mar 23 14:29:47 wazuh-server sshd[787]: Server listening on :: port 22.
Mar 23 14:29:47 wazuh-server systemd[1]: Started OpenBSD Secure Shell server.
tirthak@wazuh-server:~$ _
```

```
tirthak@wazuh-server:~$ sudo ufw status
Status: inactive
tirthak@wazuh-server:~$
```

```
PS C:\Users\Tirthak> ping 192.168.1.6

Pinging 192.168.1.6 with 32 bytes of data:
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\Tirthak>
```

```
tirthak@wazuh-server:~$ sudo ufw status
Status: inactive
tirthak@wazuh-server:~$ ss -tlnp | grep 22
LISTEN 0      128              0.0.0.0:22         0.0.0.0:*
LISTEN 0      128                 [::]:22            [::]:*
tirthak@wazuh-server:~$ _
```

```
tirthak@wazuh-server:~$ cat /etc/hosts.deny
# /etc/hosts.deny: list of hosts that are _not_ allowed to access the system.
#                  See the manual pages hosts_access(5) and hosts_options(5).
#
# Example:    ALL: some.host.name, .some.domain
#             ALL EXCEPT in.fingerd: other.host.name, .other.domain
#
# If you're going to protect the portmapper use the name "rpcbind" for the
# daemon name. See rpcbind(8) and rpc.mountd(8) for further information.
#
# The PARANOID wildcard matches any host whose name does not match its
# address.
#
# You may wish to enable this to ensure any programs that don't
# validate looked up hostnames still leave understandable logs. In past
# versions of Debian this has been the default.
# ALL: PARANOID

tirthak@wazuh-server:~$
```

```
PS C:\Users\Tirthak> Test-NetConnection -ComputerName 192.168.1.6 -Port 22
WARNING: TCP connect to (192.168.1.6 : 22) failed


ComputerName            : 192.168.1.6
RemoteAddress           : 192.168.1.6
RemotePort              : 22
InterfaceAlias          : Wi-Fi
SourceAddress           : 192.168.1.6
PingSucceeded           : True
PingReplyDetails (RTT)  : 0 ms
TcpTestSucceeded        : False



PS C:\Users\Tirthak>
```

```
  GNU nano 6.2                    /etc/netplan/00-installer-config.yaml
```

```
tirthak@wazuh-server:~$ ls /etc/netplan
50-cloud-init.yaml  99-static-ip.yaml
tirthak@wazuh-server:~$
```

```
  GNU nano 6.2
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: false
      addresses:
        - 192.168.1.6/24
      routes:
        - to: default
          via: 192.168.1.1
      nameservers:
        addresses:
          - 8.8.8.8
          - 8.8.4.4
```

Fig: - Lots of Trouble shooting and debugging because of Host machine IP changes

```
  GNU nano 6.2                          /var/ossec/etc/rules/local_rules.xml
<!-- Local rules -->

<!-- Modify it at your will. -->
<!-- Copyright (C) 2015, Wazuh Inc. -->

<!-- Example -->
<group name="local,syslog,sshd,">

  <!--
  Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2
  -->
  <rule id="100001" level="5">
    <if_sid>5716</if_sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1.</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>

</group>

                                        [ Read 19 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

```
<group name="local,windows,authentication,">

  <rule id="100002" level="10" frequency="5" timeframe="120">
    <if_matched_sid>60122</if_matched_sid>
    <description>Custom Rule: Multiple failed login attempts detected on Windows endpoint (Event ID 4625)</description>
    <mitre>
      <id>T1078</id>
    </mitre>
    <group>authentication_failed,windows,</group>
  </rule>

</group>
EOF
<!-- Local rules -->
<!-- Modify it at your will. -->
<!-- Copyright (C) 2015, Wazuh Inc. -->

<group name="local,windows,authentication,">

  <rule id="100002" level="10" frequency="5" timeframe="120">
    <if_matched_sid>60122</if_matched_sid>
    <description>Custom Rule: Multiple failed login attempts detected on Windows endpoint (Event ID 4625)</description>
    <mitre>
      <id>T1078</id>
    </mitre>
    <group>authentication_failed,windows,</group>
  </rule>

</group>
tirthak@wazuh-server:~$ |
```

Fig: - Added a Custom Detection Rule for Enhanced Threat Detection

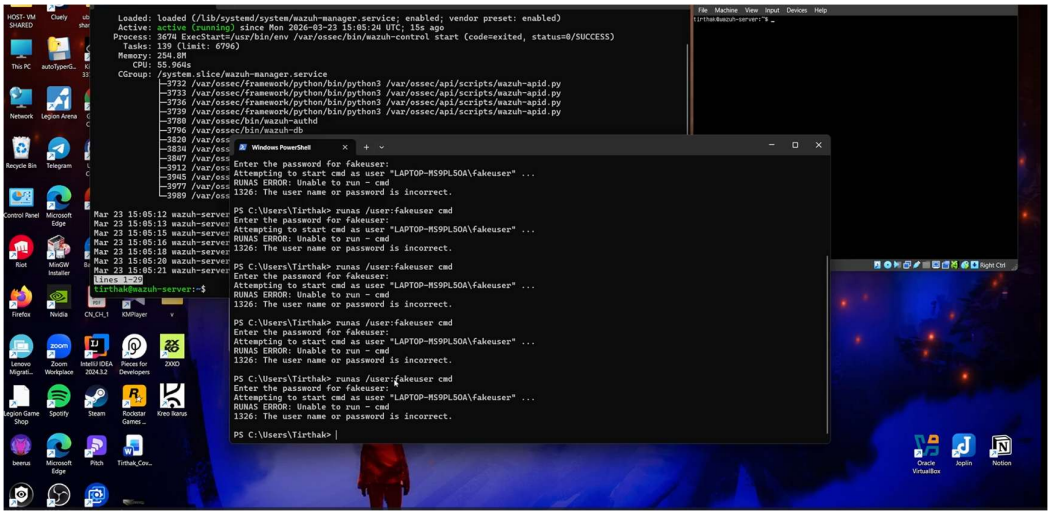
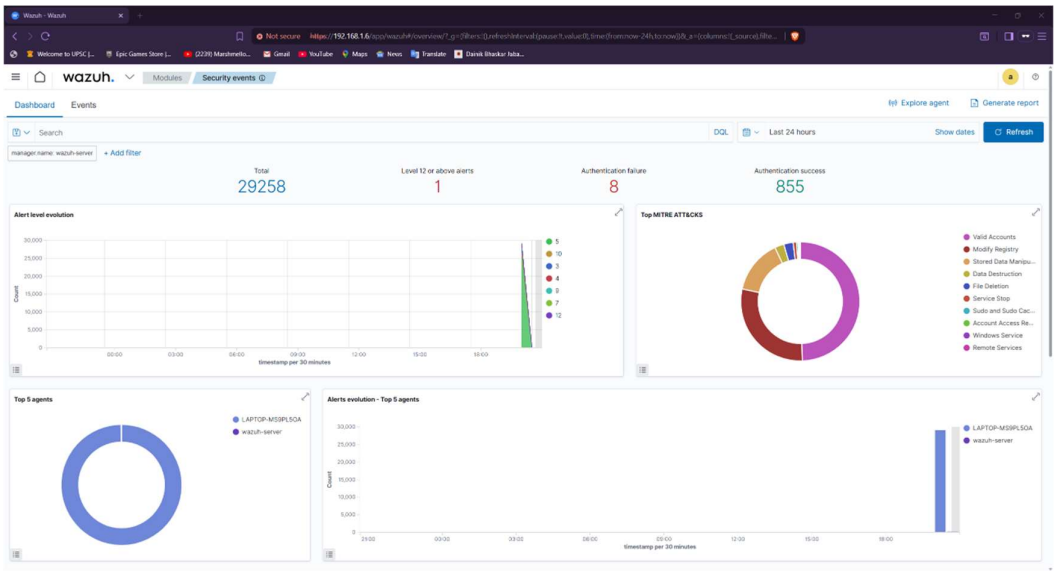
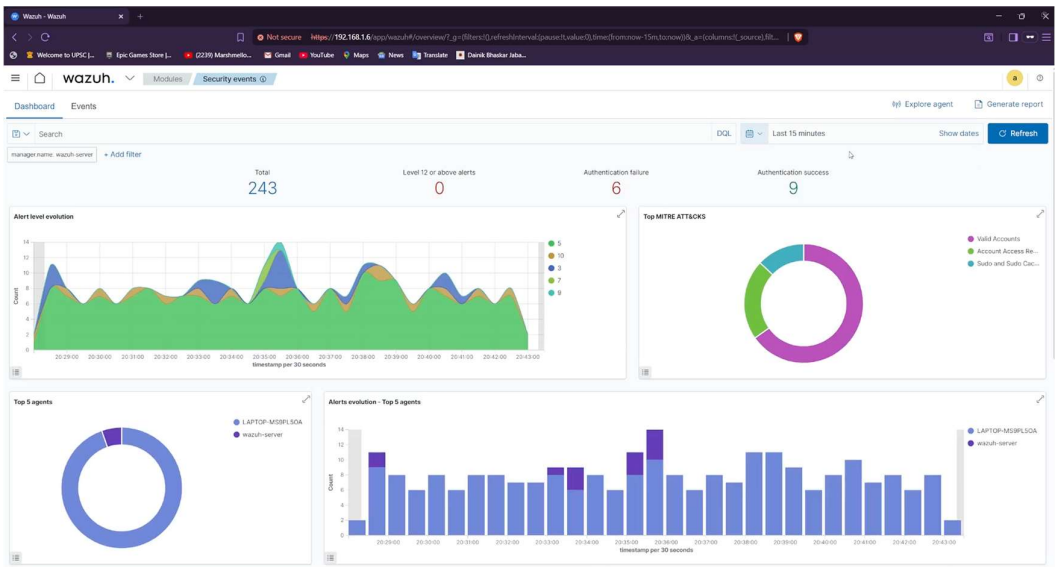


Fig: - Five and more failed login attempts detected as a Fake User





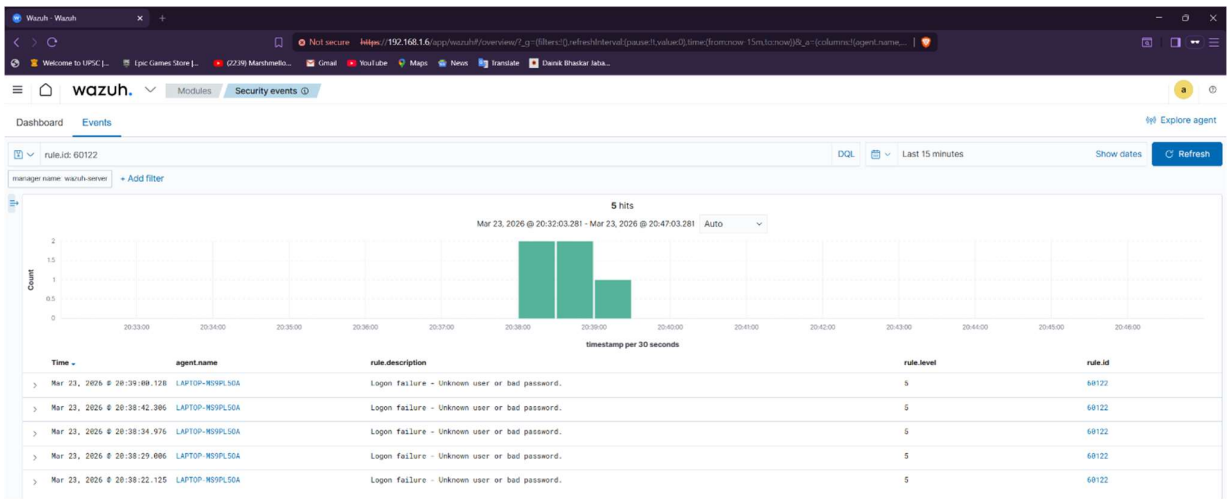
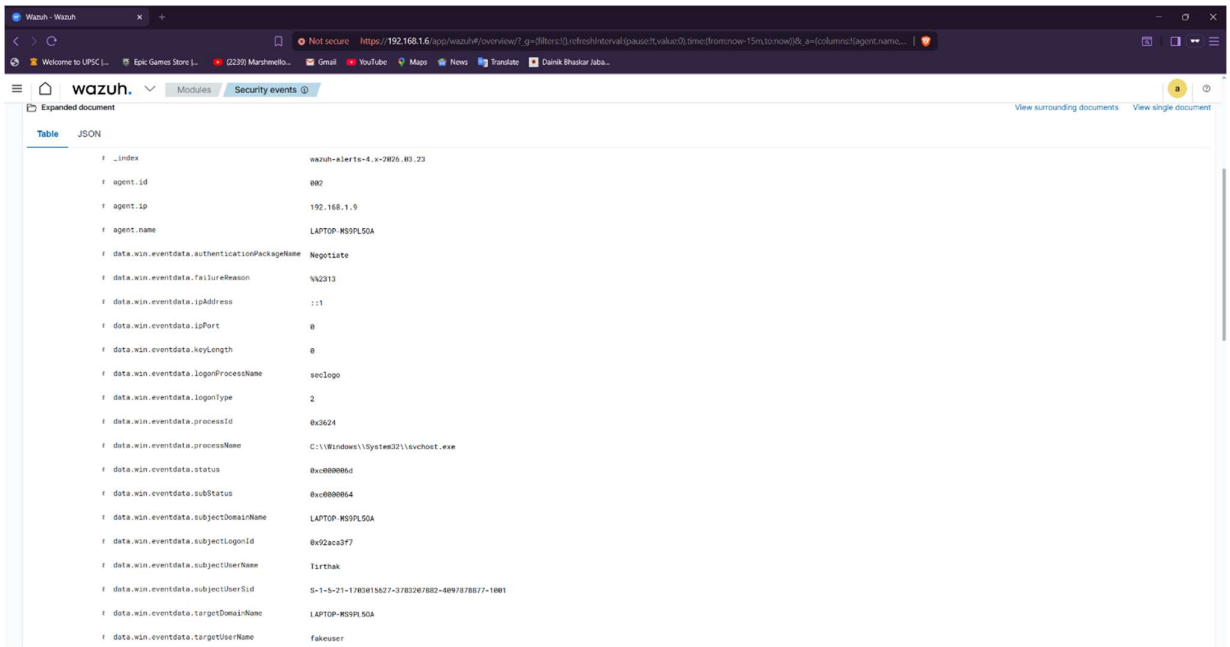
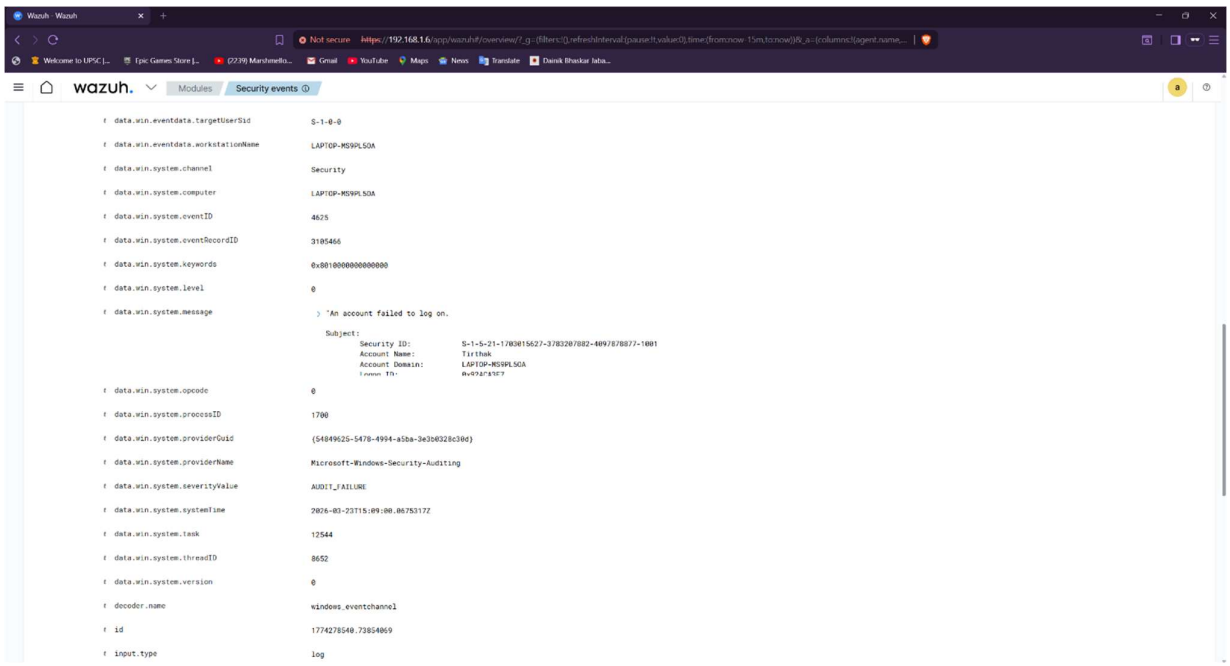
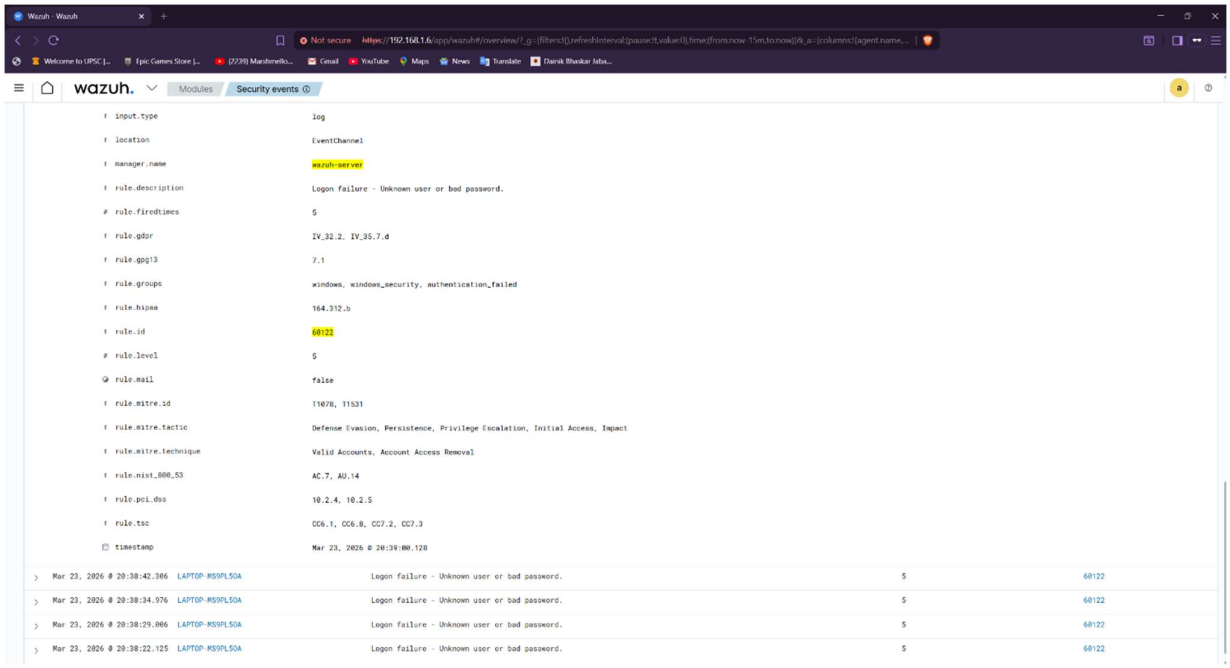


Fig: - Default Threat Detection Rule





data.win.eventdata.targetUserSid	S-1-0-0
data.win.eventdata.workstationName	LAPTOP-MS9PL5OA
data.win.system.channel	Security
data.win.system.computer	LAPTOP-MS9PL5OA
data.win.system.eventID	4625
data.win.system.eventRecordID	3105466
data.win.system.keywords	0x8010000000000000
data.win.system.level	0
data.win.system.message	"An account failed to log on. Subject: Security ID: S-1-5-21-1703015627-3783207882-4097878877-1001 Account Name: Tirthak Account Domain: LAPTOP-MS9PL5OA Logon ID: 0x97A7A5E7
data.win.system.opcode	0
data.win.system.processID	1700
data.win.system.providerGuid	{54849625-5478-4994-a5ba-3e3b0328c30d}
data.win.system.providerName	Microsoft-Windows-Security-Auditing
data.win.system.severityValue	AUDIT_FAILURE
data.win.system.systemTime	2026-03-23T15:09:00.0675317Z
data.win.system.task	12544
data.win.system.threadID	8652
data.win.system.version	0
decoder.name	windows_eventchannel
id	1774278540.73854069
input.type	log



input.type	log
location	EventChannel
manager.name	wazuh-server
rule.description	Logon failure - Unknown user or bad password.
rule.firedtimes	5
rule.gdpr	IV_32.2, IV_35.7.d
rule.gpg13	7.1
rule.groups	windows, windows_security, authentication_failed
rule.hipaa	164.312.b
rule.id	60122
rule.level	5
rule.mail	false
rule.mitre.id	T1078, T1531
rule.mitre.tactic	Defense Evasion, Persistence, Privilege Escalation, Initial Access, Impact
rule.mitre.technique	Valid Accounts, Account Access Removal
rule.nist_800_53	AC.7, AU.14
rule.pci_dss	10.2.4, 10.2.5
rule.tsc	CC6.1, CC6.8, CC7.2, CC7.3
timestamp	Mar 23, 2026 @ 20:39:00.128

Mar 23, 2026 @ 20:38:42.306	LAPTOP-MS9PL5OA	Logon failure - Unknown user or bad password.	5	60122
Mar 23, 2026 @ 20:38:34.976	LAPTOP-MS9PL5OA	Logon failure - Unknown user or bad password.	5	60122
Mar 23, 2026 @ 20:38:29.006	LAPTOP-MS9PL5OA	Logon failure - Unknown user or bad password.	5	60122
Mar 23, 2026 @ 20:38:22.125	LAPTOP-MS9PL5OA	Logon failure - Unknown user or bad password.	5	60122

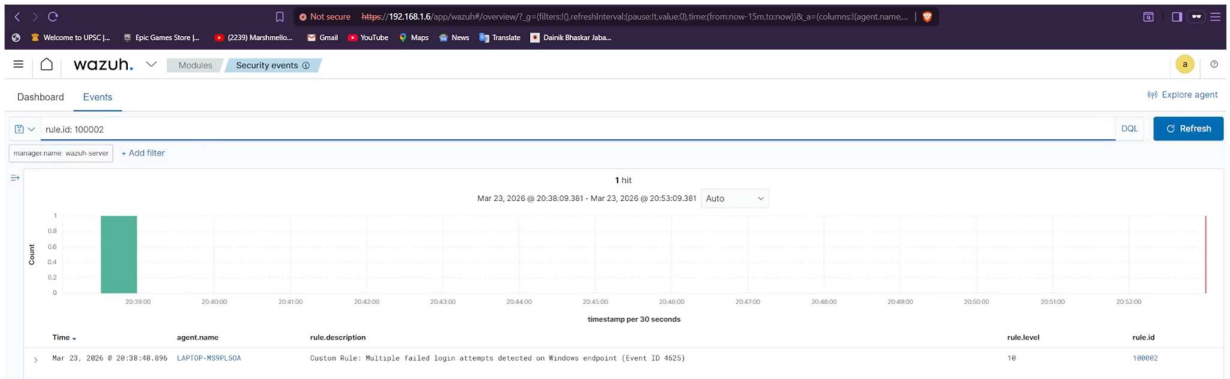


Fig: - Custom Detection Rule for Enhanced Threat Detection

Fig:- Detailed info about the detection and MITRE framework Exploitation of Defense Evasion, Privilege Exploitation and Initial Access, More Info given in the Alert Generated Table.